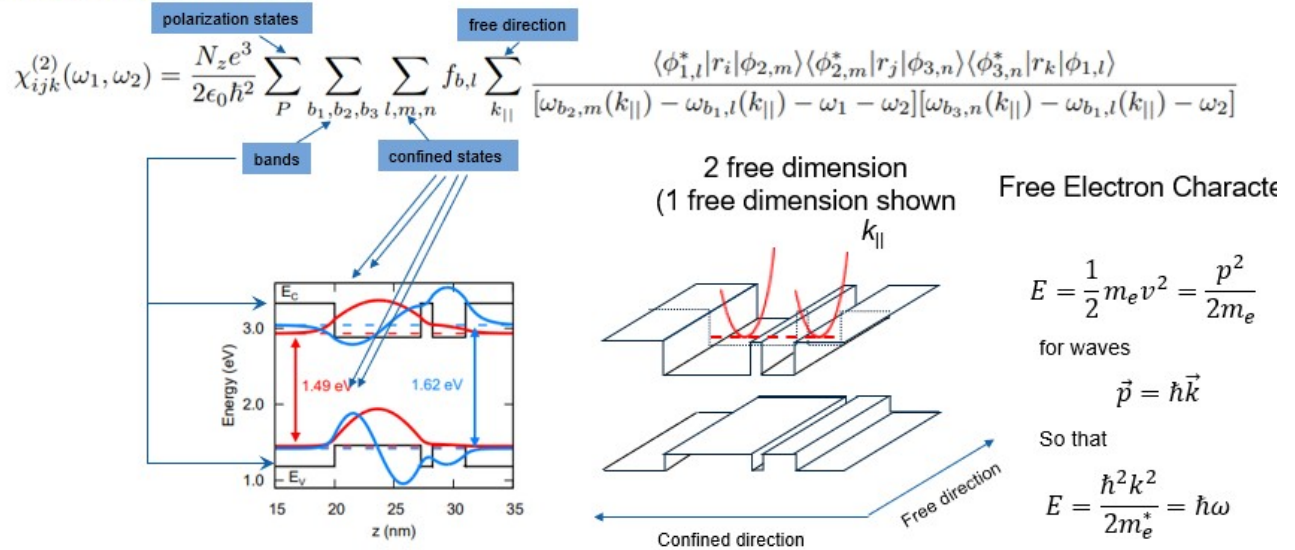


Functional Form Study Guide

A simple-to-detailed walkthrough of your boss's Equation 1 breakdown

Functional Form



Main idea: The central idea of the figure is that every quantum state in the sum needs three labels: which band it belongs to, which confined state inside that band it is, and what in-plane momentum k_{parallel} it has.

1. The whole figure in one sentence

Plain-English equation: Equation 1 calculates χ^2 by adding the contribution from every allowed optical pathway across polarization choices, bands, confined states, and in-plane momentum states.

$\chi^2 = \text{sum over polarization} \times \text{bands} \times \text{confined states} \times \text{in-plane momentum of each optical contribution}$

The three labels to remember for an electronic state are:

Label	Question it answers	Simple analogy
band b	Which band family am I in?	Which building?
state n, m, or l	Which confined level inside that band?	Which floor?
k_{parallel}	How much in-plane momentum does that state have?	How fast are you moving sideways on the floor?

Most important hierarchy: If you understand band $\rightarrow$ confined state $\rightarrow k_{\text{parallel}}$, the rest of the slide becomes much easier.

2. The equation at the top: what each summation means

$$\chi^{(2)} \sim \sum_P \sum_{(b_1, b_2, b_3)} \sum_{(l, m, n)} f_{(b, l)} \sum_{(k_{\text{parallel}})} [\text{matrix elements} / \text{resonance denominators}]$$

Sum over P: polarization states

P asks: which polarization / directional sequence of dipole interactions is being considered?

For the xzx tensor element, the optical interactions involve directions x, z, and x through operators such as r_x , r_z , and r_x .

Sum over b1, b2, b3: bands

b asks: which electronic band is the state in?

- **Conduction band:** electron-like states.
- **Heavy-hole valence band:** heavy-hole states.
- **Other possible bands:** for example light-hole or split-off bands in a more general model.

Sum over l, m, n: confined states

l, m, and n ask: which quantized state within the chosen band?

conduction band $\rightarrow e_1, e_2, e_3, \dots$ and heavy-hole band $\rightarrow hh_1, hh_2, hh_3, \dots$

Common confusion: A band and a confined state are NOT the same thing. e_1 and e_2 are two different confined states within the electron/conduction band.

3. Why the states are called "confined"

The heterostructure confines carriers in the growth direction z . The GaAs/AlGaAs layer sequence creates a potential well, so the allowed energies in z become discrete rather than continuous.

1. The layer structure creates conduction- and valence-band-edge profiles along z .
2. The Schrödinger equation is solved in this confined direction.
3. Only certain bound-state energies are allowed: e_1 , e_2 , hh_1 , hh_2 , and so on.
4. Each bound state also has an envelope wavefunction $\psi_n(z)$.

confined direction $z \rightarrow$ discrete energies E_{e1} , E_{e2} , E_{hh1} , E_{hh2} + wavefunctions $\psi(z)$

Nextnano connection: This is the part of the problem that Nextnano solves directly for the quantum-well structure.

4. Why k_{parallel} is a separate label

The carrier is confined in z , but is approximately free to move in the x - y plane of the quantum well. That means the quantum well has one confined direction and two free in-plane directions.

$$k_{\text{parallel}} = \sqrt{k_x^2 + k_y^2}$$

Direction	Behavior	Mathematical consequence
z	Confined by the heterostructure	Discrete bound states $e1$, $e2$, $hh1$, $hh2$, ...
x - y plane	Approximately free motion	Continuous in-plane momentum k_{parallel}

Critical idea: Saying "the electron is in $e1$ " is not enough to specify its full state. More completely, it is in $e1$ at a particular k_{parallel} .

$e1$ is really a subband $E_{e1}(k_{\text{parallel}})$, not one single energy point

5. What the red parabolas mean

Each confined state has an in-plane dispersion. At $k_{\text{parallel}} = 0$, the state has its minimum subband energy. As k_{parallel} increases, in-plane kinetic energy is added.

$$E_n(k_{\text{parallel}}) = E_n(0) + (\hbar^2 k_{\text{parallel}}^2)/(2 m^*)$$

The red parabolas in the figure are illustrating this free-electron-like in-plane behavior. The mass used in a semiconductor is normally an effective mass m^* , not necessarily the vacuum electron mass.

Where this comes from

1. Start with kinetic energy: $E = p^2/(2m)$.
2. Quantum mechanics relates momentum and wavevector through $p = \hbar k$.
3. Substitute $p = \hbar k$.
4. You obtain $E = \hbar^2 k^2/(2m)$.

Interpretation: The "Free Electron Character" panel is not saying the electron is free everywhere. It is saying the carrier has free-particle-like motion in the plane while remaining confined in z .

6. Why the transition energy depends on k_{parallel}

The optical transition energy is the energy difference between an electron state and a heavy-hole state at the same k_{parallel} .

$$\Delta E_{11}(k_{\text{parallel}}) = E_{e1}(k_{\text{parallel}}) - E_{hh1}(k_{\text{parallel}})$$

Because both band energies vary with k_{parallel} , the transition energy can also vary with k_{parallel} . Therefore the optical resonance condition is not determined only by the $k = 0$ energies.

Equation connection: This is why the denominator in Equation 1 contains $\omega_{b,m}(k_{\text{parallel}})$: the detuning from the laser frequency changes as you move through k -space.

At $k_{\text{parallel}} = 0$, the left plot in your boss's slide highlights transition energies around 1.49 eV and 1.62 eV. Those are examples of interband energy separations relevant to the nonlinear resonance.

7. Why Equation 1 sums over k_{parallel}

Every k_{parallel} value represents another allowed in-plane momentum state, and each can contribute a different amount to χ^2 .

$$\chi^2 = \chi^2(k_0) + \chi^2(k_1) + \chi^2(k_2) + \dots$$

So the k_{parallel} summation means: calculate the optical contribution at many in-plane momentum states and add them together.

Connection to the next step: This directly connects to your boss's previous slide, where the symbolic sum over k_{parallel} was rewritten as an explicit k -space integral.

$\text{sum}(k_{\text{parallel}}) \rightarrow \text{integral over } k_x, k_y \rightarrow \text{radial integral in } k_{\text{parallel}} \text{ when isotropy is assumed}$

8. Numerator vs denominator: what the big fraction means

Part of fraction	Plain-language question	Physical role
Matrix-element numerator	How strongly do these states couple through the three optical interactions?	Controls transition strength.
Resonance denominators	How close are the relevant transition energies to the input photon frequencies?	Can strongly enhance the response near resonance.

Mental model: A useful mental shorthand is: $\text{contribution} = \text{coupling strength} / \text{detuning from resonance}$.

What $f_{b,l}$ means

$f_{b,l}$ is an occupation factor. It asks whether the starting electronic state is occupied. In the paper's later simplified Equation 2, the relevant valence-band starting states are treated as filled, so this occupation factor becomes 1 and no longer needs to be carried explicitly.

9. How this maps to your Nextnano workflow

Concept in the figure	What you need computationally	Typical source
Band identity b	Electron or heavy-hole band	Model / state labeling
Confined state n, m, l	$e1, e2, hh1, hh2$	Nextnano eigenstates
State energy at k	$E_{e1}(k), E_{e2}(k), E_{hh1}(k), E_{hh2}(k)$	k -resolved Nextnano calculation
Envelope wavefunction	$\psi_{e,n}(z), \psi_{hh,m}(z)$	Nextnano
Transition energy	$E_{e,n}(k) - E_{hh,m}(k)$	Python post-processing
Matrix elements	overlaps and position/dipole matrix elements	Wavefunction post-processing
k_{parallel} sum/integral	Weighted accumulation over the k grid	Python numerical integration

1. Solve the coupled-QW electronic structure.
2. Track the desired $e1/e2$ and $hh1/hh2$ states across k_{parallel} .
3. Compute transition energies at each k_{parallel} .
4. Compute the matrix elements needed by the nonlinear expression.
5. Evaluate the contribution to $\chi^{(2)}$ at each k_{parallel} .
6. Apply the proper k -space weight and integrate/sum over k_{parallel} .

10. Final memorization sheet

Symbol	Say this in your head
P	Which polarization pathway?
b	Which band?
l, m, n	Which confined state inside that band?
k_{parallel}	How much sideways / in-plane momentum?
ϕ or ψ	Quantum wavefunction
r_i, r_j, r_k	Dipole / position operators for the optical interactions
$f_{b,l}$	Is the starting state occupied?
$\omega_b(k)$	Energy/frequency of that state at this k
numerator	How strongly do the states couple?
denominator	How close is the pathway to resonance?

Master key: Band $\rightarrow$ confined state $\rightarrow k_{\text{parallel}}$ is the hierarchy to memorize.

band b $\rightarrow$ confined state n $\rightarrow$ position along that state's dispersion k_{parallel}

Mastery check: If you can explain why e_1 is a subband $E_{e_1}(k_{\text{parallel}})$, rather than a single energy, you understand the most important conceptual point of your boss's slide.

Source note: This guide follows the structure and labels shown in the user-supplied "Functional Form" figure and explains those labels in the context of the coupled-quantum-well susceptibility equation discussed in this chat.